

Assignment 1: End-to-End Machine Learning Project

Loan Default Prediction Report

Dataset: Synthetic credit-risk data, 10,000 rows × 21 columns
Target: target_default_risk (0 = No Default, 1 = Default)
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1. Overview

This project predicts whether a borrower will default on a loan. It covers the full pipeline: data exploration, cleaning, feature engineering, training five classification models and hyperparameter tuning.

2. Results

Model	Accuracy	Precision	Recall	F1
XGBoost (best)	95.60%	96.44%	94.93%	95.68%
Random Forest	94.30%	96.06%	92.69%	94.35%
Logistic Regression	93.95%	96.13%	91.91%	93.97%
Decision Tree	92.80%	93.15%	92.79%	92.97%
SVM	92.70%	95.45%	90.06%	92.68%

Summary: XGBoost was the best model. Tuning improved it only slightly (95.6% → 95.8%), and Random Forest stayed at 94.3%. Income and debt_to_income were the features most related to default.

3. What Was Done

- **EDA:** data types, missing values, class balance, distributions, outliers and correlations.
- **Cleaning and features:** fixed the “Bachlors” typo, filled missing values with the median, capped outliers, added income_per_dependent and recency_days, one-hot encoded categories and scaled the data.
- **Models:** Logistic Regression, Decision Tree, SVM, Random Forest and XGBoost, evaluated with accuracy, precision, recall, F1 and confusion matrix.
- **Tuning:** GridSearchCV on Random Forest and XGBoost.

4. Tech Stack

Python 3, pandas, numpy, scikit-learn, xgboost, matplotlib, seaborn

5. Notes

- The dataset is synthetic and includes missing values, typos and outliers on purpose.
- This is a single Jupyter Notebook project, for educational purposes.

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